

SeisMamba: Selective state space models for DAS seismic denoising and phase picking

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Abstract

Distributed acoustic sensing (DAS) records 1,000–10,000 channels at sub-meter spacing, generating data volumes that require processing faster than acquisition. Existing deep learning approaches face scalability limitations: Transformers impose quadratic memory cost, recurrent networks serialize computation, and convolutional networks are bounded by their receptive field. This study proposes SeisMamba, a U-Net architecture whose sequence modeling is performed entirely by dual-axis bidirectional Mamba blocks that scan along the channel axis and the time axis at linear $\mathcal{O}(n)$ complexity. SeisMamba is benchmarked against parameter-matched ResUNet (3.98 M, convolutional) and SwinDAS (4.11 M, Swin Transformer) on two tasks: physics-based synthetic denoising and real-data P/S-wave phase picking from submarine DAS recordings. On denoising, SeisMamba achieves 26.98 ± 0.05 dB SNR, within 2.2 dB of ResUNet (29.19 ± 0.12 dB) and 7.3 dB above SwinDAS (19.70 ± 0.09 dB). On phase picking, SeisMamba attains the lowest arrival-time error ($\text{MAE}_t = 0.375 \pm 0.022$ s), 38.8% below ResUNet (0.613 s), with comparable classification accuracy across architectures. In the scalability benchmark, SeisMamba sustains approximately 28 ms inference from 1,000 to 8,000 samples and 182 ms at 65,000 samples, while ResUNet exhausts GPU memory beyond 32,000 samples. Ablation confirms that both scan axes contribute independently: removing either degrades denoising by 1.0–1.1 dB and picking accuracy by up to 2.4 percentage points. Among the tested architectures, SeisMamba offers a favorable accuracy–scalability trade-off for large-scale, continuous DAS deployments.

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Keywords: Distributed acoustic sensing, Mamba, State space model, Denoising, Phase picking, Deep learning

1. Introduction

Distributed acoustic sensing (DAS) transforms conventional fiber-optic cables into dense seismic arrays by interrogating Rayleigh backscatter along the fiber [31, 14]. A single DAS interrogator unit simultaneously records 1,000–10,000 channels at channel spacings of 1–10 m and sampling rates of 100–1,000 Hz, yielding data volumes on the order of 300 GB per day for a typical 4,000-channel, 500-Hz deployment. Arrays of this scale have proven transformative for seismic monitoring and subsurface imaging: DAS illuminates seafloor faults with dark-fiber ocean-bottom cables [14], characterizes urban site response from ambient noise [25], localizes induced seismicity in near-real time [12], and enables continuous volcano monitoring at kilometer-scale apertures [24]. The fundamental appeal of DAS lies in its spatial density: where a conventional network places one three-component seismometer per station, a DAS cable places thousands of single-component strain-rate sensors along the same path, revealing apparent-velocity and moveout information that a sparse network cannot resolve. Realizing this potential in operational monitoring, however, requires processing these massive data streams faster than they are acquired.

Deep learning has emerged as the primary algorithmic framework for DAS data processing, with published applications spanning denoising, phase picking, event detection, and ambient noise interferometry. Convolutional neural networks (CNNs) were the first architecture adopted: residual CNNs suppress fading noise in distributed vertical seismic profiling records [2], encoder-decoder networks attenuate surface-wave noise in geothermal DAS [10], and multiscale feature extractors address the heterogeneous noise environment of operational submarine arrays [8]. Self-supervised neighboring-channel prediction further extends CNN-based denoising to the label-scarce regime [29]. CNN-based phase picking [37] achieves $F_1 = 0.91$ on the Stanford DAS-N dataset by processing each channel independently and aggregating picks by weighted voting, though this design discards inter-channel coherence. Recurrent approaches exploit the sequential structure of DAS waveforms: bidirectional long short-term memory (BiLSTM) networks detect seismic events with 92% recall on traffic-monitoring arrays [13], convolutional LSTM units

model spatial-temporal coherence for volcano DAS denoising [20], and hierarchical LSTM architectures achieve 94% recall on 2,048-channel Kilauea DAS at five times real-time processing speed [24]. Attention-based Transformers have more recently been applied to exploit global context: a spatial-temporal Transformer localizes microseismic events to within 1.2 km on hydraulic-fracturing DAS arrays [27]; a BERT-style masked-prediction model picks P-wave arrivals with a mean absolute error of 8.1 ms [9]; cross-channel attention retrieves reliable Green’s functions for interferometry in seven days rather than thirty [4]. Beyond supervised approaches, self-supervised and generative methods have broadened applicability to low-label regimes: Noise2Void-style blind-spot networks [1], masked autoencoders [23], contrastive pre-training [19], and score-based diffusion models [34] all reduce dependence on manually labeled data.

Despite rapid progress, scalability remains a central challenge. Full self-attention has $\mathcal{O}(L^2)$ memory cost in sequence length L , which forces existing DAS Transformers to truncate context aggressively: Wang et al. [27] constrain their spatial-temporal Transformer to 4,096-token windows spanning only 64×64 channel-time patches; Feng et al. [4] resort to random channel subsampling; Shen et al. [23] truncate temporal sequences at 2,048 tokens. Windowed-attention variants such as the Swin Transformer reduce per-layer cost to $\mathcal{O}(L)$ by restricting attention to fixed local windows, but they sacrifice direct global context: information propagates across windows only through depth, not through a single global operation. Recurrent methods avoid the quadratic bottleneck but introduce a different constraint: sequential state transitions inhibit parallel training and limit the propagation of information across the thousands of time steps separating P and S arrivals on large-aperture arrays [7, 24]. CNNs scale linearly in input length but are bounded by their receptive field; even deep stacks struggle to capture inter-channel correlations that span hundreds of channels, the very coherence patterns that distinguish DAS from single-station recording [10, 37]. Three additional gaps follow directly from this landscape. *First*, no published work applies selective state-space models (SSMs) to DAS data: the four existing Mamba-based geophysics papers target conventional seismic denoising [11, 33], impedance inversion [28], and foundation-model pre-training [17], leaving the DAS domain entirely unaddressed. *Second*, no study provides a controlled comparison of CNN, windowed-attention Transformer, and SSM architectures on the same DAS tasks under a matched parameter budget, preventing evidence-based architecture selection for practitioners designing operational monitoring sys-

tems. *Third*, no DAS deep learning study reports inference time or memory as an explicit function of array size or sequence length, so the computational feasibility of any approach at the 10,000-channel scale remains empirically uncharacterized.

This study addresses these gaps by proposing SeisMamba, a U-Net architecture in which all sequence-modeling computations are carried out by selective state-space model (Mamba) blocks [5] rather than convolutions, LSTMs, or full self-attention. The selective scan mechanism operates in $\mathcal{O}(n)$ time and memory via a hardware-aware parallel scan algorithm [6, 5], making it tractable for the long sequences that continuous DAS monitoring demands. To exploit the joint spatial-temporal structure of DAS data, SeisMamba uses a *dual-axis bidirectional* design: one Mamba branch scans along the channel axis to capture inter-channel coherence and seismic moveout patterns, while a second branch scans along the time axis to capture waveform-level dynamics; the outputs are fused by a learnable weighted sum at each encoder level. A common backbone architecture serves both a denoising task and a P/S-wave phase-picking task through task-specific decoder heads, each trained independently on physics-based synthetic DAS data and on real field recordings, respectively.

The remainder of the paper is organized as follows. Section 2 describes the SeisMamba architecture and the comparison models. Section 3 presents the experimental setup, including synthetic data generation, real-data sources, and evaluation metrics. Section 4 reports quantitative results for denoising, phase picking, scalability, and ablation. Section 5 discusses architectural trade-offs, domain generalization, and limitations. Section 6 concludes.

2. Theory and Method

2.1. State Space Models and the Selective Scan Mechanism

Structured state space models (SSMs) provide a general framework for mapping an input sequence to an output sequence through a latent state that evolves according to linear dynamics. This section briefly introduces the formulation necessary to understand SeisMamba; readers seeking a comprehensive treatment should consult Gu et al. [6, 5].

104 *2.1.1. Continuous-time state space model*

105 The continuous-time SSM describes the latent state $h(t) \in \mathbb{R}^N$ and scalar
106 output $y(t)$ as

$$h'(t) = \mathbf{A} h(t) + \mathbf{B} x(t), \quad y(t) = \mathbf{C} h(t), \quad (1)$$

107 where $x(t) \in \mathbb{R}$ is the scalar input, $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the state transition matrix
108 that governs how information propagates through the latent state, $\mathbf{B} \in \mathbb{R}^{N \times 1}$
109 maps the current input into the state space, $\mathbf{C} \in \mathbb{R}^{1 \times N}$ projects the state
110 back onto the output, and N is the state dimension. Intuitively, $h(t)$ is a
111 compressed memory of all past inputs; the choice of $\mathbf{A}$ determines how quickly
112 this memory decays and which input patterns are selectively retained.

113 *2.1.2. Discrete-time formulation via zero-order hold*

114 To process sampled sequences, the continuous-time parameters $(\mathbf{A}, \mathbf{B})$
115 are discretized using the zero-order hold (ZOH) rule with a learnable step
116 size $\Delta > 0$:

$$\bar{\mathbf{A}} = \exp(\Delta \mathbf{A}), \quad \bar{\mathbf{B}} = (\Delta \mathbf{A})^{-1}(\exp(\Delta \mathbf{A}) - \mathbf{I}) \cdot \Delta \mathbf{B}. \quad (2)$$

117 The resulting discrete recurrence is

$$h_k = \bar{\mathbf{A}} h_{k-1} + \bar{\mathbf{B}} x_k, \quad y_k = \mathbf{C} h_k, \quad (3)$$

118 where k indexes the discrete time step. Equation (3) has the form of a linear
119 recurrence, but it can equivalently be expressed as a global convolution with
120 kernel $\mathbf{C} \bar{\mathbf{A}}^k \bar{\mathbf{B}}$, enabling parallel computation during training via fast Fourier
121 transform [6].

122 *2.1.3. Mamba’s selective scan (S6)*

123 The S4 model [6] uses input-independent matrices $(\mathbf{A}, \mathbf{B}, \mathbf{C}, \Delta)$, applying
124 the same filtering operation regardless of input content. Mamba [5] overcomes
125 this limitation through the *selective scan* mechanism, denoted S6, in which
126 $\mathbf{B}$, $\mathbf{C}$, and Δ become functions of the current input token x_k :

$$B_k = s_B(x_k), \quad C_k = s_C(x_k), \quad \Delta_k = \text{softplus}(s_\Delta(x_k)), \quad (4)$$

127 where s_B , s_C , and s_Δ are learned linear projections. This *content-based*
128 *gating* allows the model to selectively retain or discard information at each

position: a large Δ_k contracts $\bar{\mathbf{A}}$ toward zero, forcing the latent state to reset; a small Δ_k preserves the accumulated state across many steps. In the DAS context this behavior is physically meaningful—the model can suppress ambient-noise intervals by resetting its state while propagating information continuously across the tens of seconds separating P- and S-wave arrivals on a large-aperture array. Because selectivity makes the system time-varying, the efficient convolution shortcut of S4 is no longer applicable; instead, Mamba employs a hardware-aware parallel scan algorithm that computes the full recurrence in $\mathcal{O}(n)$ time and memory by fusing operations within GPU shared memory [5]. This linear complexity is the central computational advantage of Mamba over Transformer self-attention ($\mathcal{O}(n^2)$) and over recurrent LSTM whose sequential state transitions prevent batch-parallel training [7]. The Mamba 2 formulation [3] further establishes a formal duality between structured SSMS and linear attention, providing additional theoretical grounding for the linear-complexity claim.

2.2. SeisMamba Architecture

SeisMamba is a hierarchical encoder-decoder (U-Net style) network in which all sequence-modeling computations are performed by DualMamba blocks—the novel architectural unit introduced in this work. Figure 1 illustrates the overall design.

The network accepts a single-channel DAS gather of shape $(N_c \times N_t)$, where N_c is the number of channels and N_t is the number of time samples, and produces either a denoised gather of the same shape (denoising task) or per-sample three-class phase probabilities (picking task).

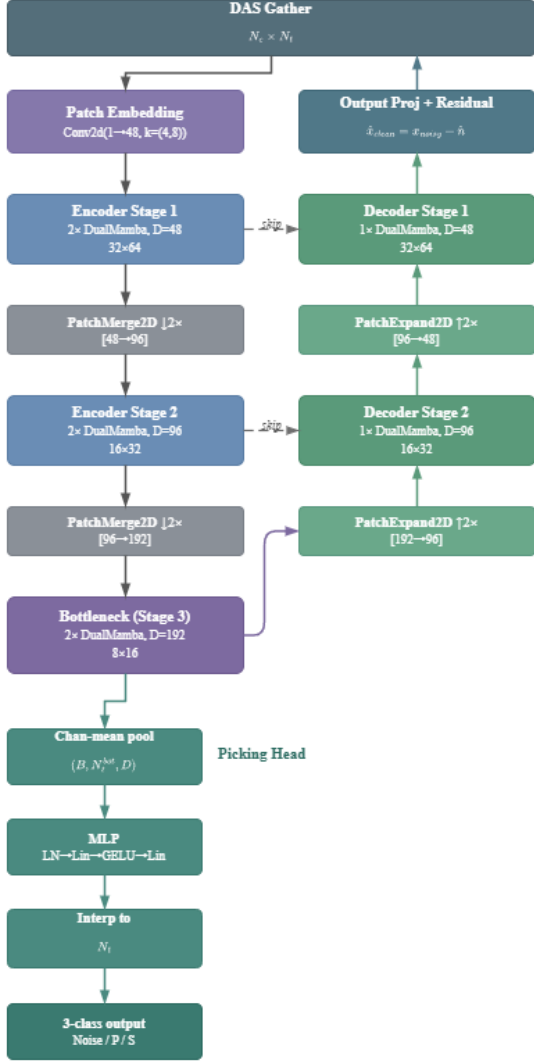
2.2.1. Patch embedding

Raw DAS data are tokenized by a strided 2D convolution:

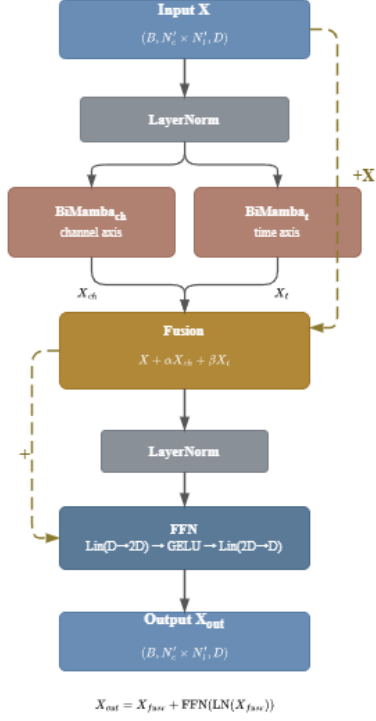
$$\mathbf{E} = \text{Conv2d}(1 \rightarrow D_0, \text{kernel} = (4, 8), \text{stride} = (4, 8)), \quad (5)$$

with base feature dimension $D_0 = 48$. The asymmetric kernel reflects the physical scales of DAS data: a factor of 4 along the channel axis corresponds to 40 m of fiber at a 10 m channel spacing, a distance over which apparent slowness varies smoothly; a factor of 8 along the time axis corresponds to 80 ms at 100 Hz sampling, comfortably above the Nyquist period of the dominant seismic frequency band below 10 Hz. This asymmetric tokenization respects the anisotropic information density of DAS gathers and distinguishes

(a) SeisMamba Architecture



(b) DualMamba Block



(c) BiMamba Unit

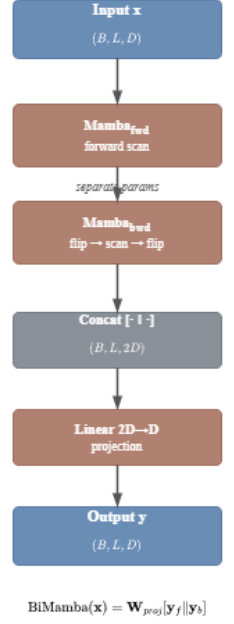


Figure 1: SeisMamba architecture. (a) U-Net encoder–decoder with denoising and picking heads. (b) DualMamba block with parallel channel- and time-axis BiMamba branches. (c) BiMamba unit with bidirectional Mamba scans.

162 SeisMamba from image-processing Mamba models that assume square spatial
 163 patches [35, 15].

164 The output token grid has shape $(N_c/4, N_t/8, D_0)$. A decomposed 2D
 165 positional embedding is then added:

$$\mathbf{E}_{\text{pos}} = \mathbf{E} + \mathbf{p}_{\text{ch}} \oplus \mathbf{p}_{\text{t}}, \quad (6)$$

166 where $\mathbf{p}_{\text{ch}} \in \mathbb{R}^{(N_c/4) \times D_0}$ and $\mathbf{p}_{\text{t}} \in \mathbb{R}^{(N_t/8) \times D_0}$ are learnable vectors broadcast-
 167 added along their respective axes. Factorizing the positional embedding into
 168 independent channel and time components reduces the number of learnable
 169 parameters and allows the model to generalize to gather shapes not seen
 170 during training.

171 2.2.2. DualMamba block

172 The DualMamba block is the basic processing unit of SeisMamba. Each
 173 block receives a feature tensor $\mathbf{X} \in \mathbb{R}^{B \times N'_c \times N'_t \times D}$ and applies a pre-normalization
 174 step $\hat{\mathbf{X}} = \text{LayerNorm}(\mathbf{X})$ before routing through two parallel BiMamba
 175 branches and fusing the results with a learnable weighted sum.

176 *BiMamba..* A BiMamba unit applies bidirectional Mamba scanning along
 177 a designated sequence axis. The forward branch applies a Mamba module
 178 ($d_{\text{state}} = 16, d_{\text{conv}} = 4, \text{expand} = 2$) to the sequence in its natural order;
 179 the backward branch flips the sequence, applies a *separate* Mamba module
 180 with independent parameters, and flips the output back. The two outputs
 181 are concatenated along the feature axis and projected to dimension D by a
 182 linear layer:

$$\text{BiMamba}(\mathbf{x}) = \mathbf{W}_{\text{proj}}[\text{Mamba}_{\text{fwd}}(\mathbf{x}) \parallel \text{flip}(\text{Mamba}_{\text{bwd}}(\text{flip}(\mathbf{x})))], \quad (7)$$

183 where $[\cdot \parallel \cdot]$ denotes concatenation along the feature dimension. Bidirectional-
 184 ity is essential for DAS data because seismic wavefields are non-causal along
 185 the fiber: waves arriving from either end of the array produce coherent move-
 186 out in opposite directions, and a unidirectional scan would impose an artificial
 187 asymmetry on the learned representations. This bidirectional design follows
 188 the precedent established by Vision Mamba for 2D image sequences [35] and
 189 U-Mamba for biomedical image segmentation [18].

190 *Channel-axis BiMamba..* The first branch captures inter-channel dependencies—
 191 apparent velocity, seismic moveout, and spatial coherence patterns across the

192 fiber aperture. The feature tensor is reshaped to (BN'_t, N'_c, D) , treating each
 193 time-slice as an independent batch element, and BiMamba is applied along
 194 the N'_c -length channel axis:

$$\mathbf{X}_{\text{ch}} = \text{reshape}(\text{BiMamba}_{\text{ch}}(\text{reshape}(\hat{\mathbf{X}}, BN'_t \times N'_c \times D)), B \times N'_c \times N'_t \times D). \quad (8)$$

195 Scanning the full channel axis in a single pass allows the model to propagate
 196 slowness information across the entire array aperture—a global operation
 197 that finite-receptive-field CNNs can only approximate through deep stacking
 198 [10].

199 *Time-axis BiMamba..* The second branch captures temporal dynamics within
 200 each channel—waveform shape, frequency content, and the temporal rela-
 201 tionship between P- and S-wave arrivals. The tensor is reshaped to (BN'_c, N'_t, D)
 202 and BiMamba is applied along the N'_t -length time axis:

$$\mathbf{X}_{\text{t}} = \text{reshape}(\text{BiMamba}_{\text{t}}(\text{reshape}(\hat{\mathbf{X}}, BN'_c \times N'_t \times D)), B \times N'_c \times N'_t \times D). \quad (9)$$

203 Because the time axis is scanned with $\mathcal{O}(N'_t)$ parallel complexity, this branch
 204 achieves the long-range temporal modeling capability of LSTMs [7] while
 205 supporting full batch parallelism.

206 *Learnable axis fusion..* The outputs of the two BiMamba branches are com-
 207 bined by a learnable weighted residual sum:

$$\mathbf{X}_{\text{fuse}} = \mathbf{X} + \alpha \mathbf{X}_{\text{ch}} + \beta \mathbf{X}_{\text{t}}, \quad (10)$$

208 where α and β are scalar learnable parameters initialized to 0.5. Equal
 209 initialization imposes no a priori preference for either axis; the model adapts
 210 the balance to the task during training. Ablation experiments in Section 4
 211 confirm that both axes contribute positively and that their joint contribution
 212 exceeds either branch alone.

213 *Feed-forward network..* The fused tensor is passed through a position-wise
 214 feed-forward network:

$$\mathbf{X}_{\text{out}} = \mathbf{X}_{\text{fuse}} + \text{Linear}_{2D \rightarrow D}(\text{GELU}(\text{Linear}_{D \rightarrow 2D}(\text{LayerNorm}(\mathbf{X}_{\text{fuse}})))). \quad (11)$$

215 This FFN applies a channel-mixing transformation independently at each
 216 spatial position, analogous to the FFN sublayer in standard Transformer
 217 blocks.

218 2.2.3. Hierarchical encoder

219 The encoder consists of three stages with feature dimensions [48, 96, 192].
 220 Each stage applies two successive DualMamba blocks; between stages a
 221 PatchMerge2D layer halves the spatial resolution in both axes while dou-
 222 bling the feature dimension:

$$\text{PatchMerge2D} = \text{LayerNorm}(\text{Conv2d}(D_s \rightarrow D_{s+1}, \text{kernel} = 2, \text{stride} = 2)). \quad (12)$$

223 Progressive downsampling serves two complementary purposes: it reduces
 224 the token-sequence length processed at deeper stages, lowering the cost of
 225 the global BiMamba scans, and it builds a multiscale feature hierarchy from
 226 which the decoder can reconstruct fine spatial detail through skip connec-
 227 tions. After three downsampling steps the encoder bottleneck operates at
 228 resolution $(N_c/32, N_t/64)$ with $D = 192$ features; skip-connection feature
 229 maps from each encoder stage are stored for the decoder.

230 2.2.4. Task-specific decoders

231 *Denoising decoder.* The decoder mirrors the encoder symmetrically. At each
 232 stage, a transposed convolution (stride = 2) upsamples the feature map; the
 233 upsampled features are concatenated with the corresponding encoder skip
 234 connection and reduced back to the stage dimension by a linear projection;
 235 a single DualMamba block then refines the combined representation. The
 236 output projection reshapes tokens to pixel space through two linear layers
 237 separated by a GELU activation.

238 Because the denoising target (clean signal) differs from the input (noisy
 239 signal) by a small noise component, SeisMamba employs *residual learning*:
 240 the decoder predicts the noise $\hat{n}$, and the denoised estimate is formed as

$$\hat{x}_{\text{clean}} = x_{\text{noisy}} - \hat{n}. \quad (13)$$

241 This formulation, introduced for image denoising by DnCNN [32] and sub-
 242 sequently adopted for DAS fading noise suppression [2], biases the network
 243 toward predicting small perturbations rather than full waveforms, which ac-
 244 celerates convergence and improves fidelity at high signal-to-noise ratios. All
 245 three task-benchmark models (SeisMamba, ResUNet, SwinDAS) use this
 246 identical residual-learning convention to ensure that any observed perfor-
 247 mance difference reflects architectural capacity rather than the choice of out-
 248 put parameterization.

249 *Phase-picking head.* The phase-picking head branches directly from the en-
250 coder bottleneck, bypassing the reconstruction decoder. Channel-axis mean
251 pooling collapses the spatial dimension to yield a time-sequence represen-
252 tation of shape (B, N_t^{bot}, D) , where $N_t^{\text{bot}} = N_t/64$ at the bottleneck. A
253 two-layer MLP (LayerNorm $\rightarrow$ Linear $\rightarrow$ GELU $\rightarrow$ Linear) maps each bot-
254 tleneck token to a three-dimensional logit vector for the classes {Noise, P-
255 wave, S-wave}. The logits are then upsampled to the original time resolution
256 N_t by linear interpolation, producing per-sample predictions $\hat{\mathbf{p}} \in \mathbb{R}^{B \times N_t \times 3}$.
257 Pooling over channels before classification encodes the physically motivated
258 assumption that seismic arrivals manifest coherently across many channels;
259 this aggregation integrates spatial evidence without requiring the picking
260 head to resolve single-channel noise realizations. This assumption is appro-
261 priate for near-vertical arrivals from distant sources, where the onset time is
262 approximately constant across the 128-channel patch (1.28 km aperture); for
263 wide-angle DAS configurations with significant moveout across the patch, a
264 spatially resolved picking head would be required.

265 2.2.5. Total parameter count and computational complexity

266 SeisMamba has 4.16 million trainable parameters. Each DualMamba
267 block performs two BiMamba scans, each invoking the $\mathcal{O}(n)$ parallel scan
268 algorithm, so inference time grows linearly with the number of tokens $n =$
269 $N_c' \times N_t'$. Table 1 collects the architecture configurations and complexity
270 characteristics of all models evaluated in this study.

271 2.3. Comparison Architectures

272 To isolate the contribution of the SSM design, SeisMamba is compared
273 against two full-featured DAS processing networks trained under an identical
274 parameter budget and optimization protocol. A third model, BiLSTM-DAS,
275 participates only in the scalability benchmark; it is excluded from the task
276 benchmarks because its sequential processing imposes a training-time over-
277 head exceeding 30 times that of the other models at the dataset sizes used
278 here, making a fair comparison within the available computational budget
279 impractical.

280 2.3.1. ResUNet

281 ResUNet is a four-stage encoder-decoder with residual convolutional blocks,
282 providing the CNN baseline. Each encoder stage applies two residual blocks
283 of the form Conv3 $\times$ 3 $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Conv3 $\times$ 3 $\rightarrow$ BatchNorm

Table 1: Architecture configurations and computational complexity of all compared models. All models target approximately 4 million parameters for a controlled comparison. D_0 denotes the base feature dimension; stage dims lists the feature dimension at each encoder stage.

Model	Architecture type	Parameters	Stage dims	Complexity
SeisMamba	Dual-axis Bi-Mamba U-Net	4.16M	[48, 96, 192]	$\mathcal{O}(n)$
ResUNet	Residual CNN U-Net	3.98M	[32, 64, 128, 256]	$\mathcal{O}(n)$
SwinDAS	Swin Transformer U-Net	4.11M	[56, 112, 224]	$\mathcal{O}(n)^a$
BiLSTM-DAS	Bidirectional LSTM	3.88M	hidden = 256	$\mathcal{O}(n)^b$

^a $\mathcal{O}(n)$ per layer within windows of fixed size 8×16 tokens; global context beyond the window relies on depth rather than a single global scan.

^b Algorithmic complexity is $\mathcal{O}(n)$ per layer, but sequential state updates prevent GPU parallelism along the time axis; wall-clock cost grows linearly with a large constant, making BiLSTM orders of magnitude slower than convolutional and SSM models in practice.

284 with an additive skip connection, followed by MaxPool2d (2×2) downsam-
285 pling. Stage feature dimensions progress as [32, 64, 128, 256]; the decoder
286 mirrors this hierarchy with transposed convolutions and skip-connection con-
287 catenations. At 3.98 million parameters, ResUNet characterizes what a well-
288 tuned convolutional network achieves under a comparable parameter budget
289 and the same training protocol. Because each convolution operates within a
290 fixed receptive field, the model’s ability to capture inter-channel coherence
291 across the full array aperture is limited by network depth; compensating
292 through deeper stacking would require substantially more parameters [10, 8].

293 2.3.2. SwinDAS

294 SwinDAS is a hierarchical U-Net built on the Swin Transformer [27],
295 adapted for DAS data. The encoder uses embedding dimension 56 with stage
296 depths [2, 4, 4] and attention heads [2, 4, 8], yielding stage feature dimensions
297 [56, 112, 224]. Between stages, PatchMerging (2×2 spatial concatenation fol-
298 lowed by a linear projection) halves the spatial resolution and doubles the
299 feature width. The decoder applies PatchExpanding (a linear projection re-
300 arranged by pixel-shuffle) for upsampling, with skip connections from each

301 encoder stage.

302 The attention windows are asymmetric (8 channel tokens $\times$ 16 time to-
303 kens), matching the aspect ratio of DAS gathers: the channel direction spans
304 tens of meters per token while the time direction encodes hundreds of millisec-
305 onds per token. Alternating W-MSA and SW-MSA layers with a cyclic-shift
306 masking scheme enable cross-window information exchange, and per-head
307 relative position biases are learned within each window.

308 SwinDAS has 4.11 million parameters and represents the Transformer
309 paradigm. Critically, windowed attention achieves $\mathcal{O}(n)$ total complexity
310 per layer—the same asymptotic scaling as SeisMamba—but restricts direct
311 attention to tokens within a fixed 8×16 window. Context beyond this win-
312 dow accumulates gradually through the shifted-window mechanism and hier-
313 archical downsampling rather than through a single global state transition.
314 Comparing SwinDAS and SeisMamba therefore tests two $\mathcal{O}(n)$ architectures
315 that differ in *how* global context is built: SeisMamba by scanning the full se-
316 quence at each block via the SSM recurrence, SwinDAS by attending locally
317 and propagating context through depth.

318 2.3.3. BiLSTM-DAS

319 BiLSTM-DAS comprises a spatial convolution front-end followed by three
320 layers of bidirectional LSTM [7] with hidden dimension 256 per direction, to-
321 taling 3.88 million parameters. Despite the matched parameter budget, the
322 model’s sequential recurrence renders its GPU utilization qualitatively dif-
323 ferent from the other architectures: at a time length of $N_t = 8,000$ samples,
324 BiLSTM-DAS requires approximately 29 seconds of GPU time per batch
325 compared with 28 ms for SeisMamba—a factor of roughly 1,000. This over-
326 head arises because each LSTM time step must receive the hidden state of
327 the previous step before it can begin computation, preventing the GPU from
328 exploiting spatial parallelism within the batch. BiLSTM-DAS is included
329 in the scalability benchmark to quantify this overhead empirically and to
330 provide a reference that reflects the de facto standard for sequential DAS
331 processing [13, 24].

332 2.4. Training Protocol

333 All three task-benchmark models (SeisMamba, ResUNet, SwinDAS) are
334 trained under identical optimization conditions to ensure that any observed
335 performance difference reflects architectural design rather than training-procedure
336 bias. The AdamW optimizer is used with an initial learning rate of 3×10^{-4} , a

cosine annealing schedule over the full training duration, and a linear warm-up over the first 5 epochs. The batch size is 8 for all models. Each model is trained with three independent random seeds (42, 123, 456); results are reported as mean $\pm$ standard deviation across seeds to quantify run-to-run variability. Complete details of the loss functions, evaluation metrics, data sources, and training durations are given in Section 3.

3. Experimental Setup

3.1. Synthetic DAS Data for Denoising

Training data were generated by propagating Ricker wavelets (5–30 Hz) through four submarine velocity model families using the acoustic wave simulator Deepwave [22]: (A) layered (3 sediment layers, 1600–3500 m/s); (B) gradient (linear compaction from 1600 to 4000 m/s); (C) complex (8–12 laterally heterogeneous layers, Marmousi-like, 1600–4500 m/s); and (D) stochastic (von Karman perturbation, ± 5 –10%). All models include a 200–500 m water column at 1500 m/s. This physics-based approach ensures realistic propagation effects—wavefront curvature, multiples, moveout—in contrast to plane-wave surrogates [10, 29].

Receivers form a 128-channel horizontal array at 10 m spacing along the seafloor; sources are placed at randomized sub-seafloor depths. A total of 250 shots per model family yield 1,000 gathers. Deepwave displacement output is converted to DAS strain rate by spatial differentiation (gauge length $g = 10$ m), temporal differentiation, gauge-length averaging, and decimation to 100 Hz (512 samples, 5.12 s record) [31]. Each gather is then zero-meanded and unit-variance normalized to produce $x_{\text{clean}} \in \mathbb{R}^{128 \times 512}$.

Four noise types cover the principal DAS interference sources (Table 2): Gaussian white, colored ($f^{-\alpha}$, $\alpha \sim \mathcal{U}[0.5, 2.0]$), coherent moveout (multiple linear-moveout events with varying apparent velocities 200–3000 m/s, frequencies 2–25 Hz, and random onset times), and mixed (random combination of two or three of the preceding types). For each gather one type is selected uniformly at random (25% each), scaled to a target SNR drawn from the ranges in Table 2, and added to x_{clean} .

The 1,000 gathers are split 800/100/100 (train/validation/test) by stratified sampling with equal representation from each velocity family. The test set is held out from all training and hyperparameter decisions.

Table 2: Noise models used in synthetic DAS denoising training. SNR is defined as $10 \log_{10}(\|x_{\text{clean}}\|^2 / \|x_{\text{noise}}\|^2)$.

Type	Description	SNR range (dB)
Gaussian white	Broadband random noise; baseline for comparison	3–20
Colored ($f^{-\alpha}$)	Low-frequency ocean ambient noise with spectral slope $\alpha \sim \mathcal{U}[0.5, 2.0]$	5–15
Coherent	Multiple linear-moveout events (4–8) with varying apparent velocities and frequencies, mimicking ship noise and water-column interference	5–15
Mixed	Random combination of two or three of the above types	3–15

3.2. Real DAS Data for Phase Picking

Phase picking requires per-pixel class labels (Noise, P-wave, S-wave) for each time sample. Acoustic wave simulation can produce P-wave arrivals but cannot generate S-wave phases on a fluid-only model; using synthetic labels would therefore produce a degenerate two-class problem. Instead, the publicly available DeepSubDAS dataset is used, which provides human-annotated pixel-level labels on real submarine DAS records [30].

The DeepSubDAS dataset comprises 74 labeled patches, each approximately $300 \text{ channels} \times 4000 \text{ time samples}$, recorded on a submarine fiber-optic cable in a tectonically active region. Three-class pixel annotations—Noise, P-wave, and S-wave—were assigned by domain experts.

Because all models expect a fixed input size of 128×512 , each training iteration extracts a random crop of that shape from the full 300×4000 patch: a random top-left corner (c_0, t_0) is sampled uniformly within the valid range, and the sub-array $[c_0 : c_0+128, t_0 : t_0+512]$ is extracted for both the DAS data and the corresponding pixel-level label mask. Two independent crops are drawn per file per epoch (`patches_per_file=2`), yielding $59 \times 2 = 118$ training samples per epoch. Each crop is zero-mean, unit-variance normalized using the statistics of that crop alone. The per-pixel label array of shape $(N_c, N_t, 3)$ is collapsed to $(N_t, 3)$ by channel-mean averaging of the class

391 probability maps before loss computation, because the SeisMamba picking
 392 head produces time-axis predictions (one label vector per time sample rather
 393 than per channel–time pixel).

394 Files are divided 80/20 by recording segment (59 training files and 15
 395 validation files) to prevent temporal leakage between splits. The 15 validation
 396 files also serve as the evaluation set for final metric reporting; no additional
 397 held-out test partition is created because the small dataset size (74 files total)
 398 precludes a three-way split without substantially reducing training data.

399 Class imbalance is severe: P-wave arrivals occupy 4.6–22.2% of samples
 400 across files (median 8.3%), and S-wave arrivals occupy a similar fraction; the
 401 remaining samples are labeled Noise. This imbalance motivates the weighted
 402 cross-entropy loss described in Section 3.3.

403 3.3. Loss Functions

404 The denoising objective combines MSE with a structural similarity penalty:

$$\mathcal{L}_{\text{denoise}} = \text{MSE}(\hat{x}, x_{\text{clean}}) + 0.1 \times (1 - \text{SSIM}(\hat{x}, x_{\text{clean}})), \quad (14)$$

405 where $\hat{x} = x_{\text{noisy}} - \hat{n}$ is the denoised estimate (Equation (13)) and SSIM
 406 uses a 7×7 Gaussian window ($\sigma = 1.5$); evaluation SSIM uses the standard
 407 11×11 window for comparability with published benchmarks. The weight
 408 0.1 was selected from $\{0.01, 0.05, 0.1, 0.2\}$ by validation-set SNR.

409 Phase picking uses class-weighted cross-entropy:

$$\mathcal{L}_{\text{pick}} = - \sum_{t=1}^{N_t} \sum_{c \in \{\mathcal{N}, \mathcal{P}, \mathcal{S}\}} w_c y_{t,c} \log \hat{p}_{t,c}, \quad (15)$$

410 with weights $\mathbf{w} = [0.05, 1.0, 1.0]$ for $[\mathcal{N}, \mathcal{P}, \mathcal{S}]$. The low Noise weight
 411 prevents a degenerate all-Noise solution that would achieve high accuracy
 412 while detecting no arrivals.

413 3.4. Evaluation Metrics

414 Denoising quality is measured by three metrics: signal-to-noise ratio

$$\text{SNR}(\hat{x}, x_{\text{clean}}) = 10 \log_{10} \frac{\|x_{\text{clean}}\|^2}{\|x_{\text{clean}} - \hat{x}\|^2}, \quad (16)$$

415 structural similarity (SSIM, 11×11 window), and Pearson correlation (Corr),
 416 capturing energy fidelity, structural preservation, and waveform coherence,
 417 respectively [2].

418 Phase picking is evaluated by per-class F1 scores ($F1_P$, $F1_S$), pixel-level
 419 accuracy (Acc), and arrival-time mean absolute error:

$$\text{MAE}_t = \frac{1}{M} \sum_{m=1}^M |\hat{t}_m^{\mathcal{P}} - t_m^{\mathcal{P}}|, \quad (17)$$

420 where $\hat{t}_m^{\mathcal{P}}$ is the time of peak P-wave softmax probability and $t_m^{\mathcal{P}}$ is the anno-
 421 tated onset. MAE_t is more diagnostic of operational picking precision than
 422 F1 alone because it penalizes broad, temporally diffuse detections.

423 Scalability is assessed by inference time (ms) and peak GPU memory
 424 (MB) on a single NVIDIA RTX 4090 (24 GB) with batch size 1. Ten warm-
 425 up passes precede 50 timed passes; the median is reported. Configurations
 426 causing OOM are excluded from further timing.

427 3.5. Training Details

428 Table 3 lists the shared hyperparameters. All models use AdamW [16]
 429 with cosine annealing and early stopping (patience 15). Each model-task
 430 combination is repeated with three seeds; results are reported as mean $\pm$ std.
 431 Online augmentation (time shift, channel flip, amplitude scaling) is applied
 432 during denoising training only; picking labels are not augmented because the
 433 P/S-wave annotation structure must be preserved.

434 3.6. Scalability Benchmark Protocol

435 Two sweeps measure inference throughput on random $\mathcal{N}(0, 1)$ tensors
 436 of shape $(1, 1, N_c, N_t)$. The *sequence-length sweep* fixes $N_c = 256$ and varies
 437 $N_t \in \{1024, \dots, 65536\}$ (up to 10.9 min of 100 Hz record). The *channel-count*
 438 *sweep* fixes $N_t = 512$ and varies $N_c \in \{128, \dots, 8192\}$ [26]. All four models
 439 participate; 10 warm-up passes precede 50 timed passes (`cudnn.benchmark`
 440 `= True`), and the median elapsed time is reported.

441 4. Results

442 The results are organized into four subsections. Section 4.1 evaluates
 443 denoising performance on the held-out synthetic test set. Section 4.2 reports
 444 phase-picking accuracy and arrival-time precision on the real DeepSubDAS
 445 data. Section 4.3 quantifies computational scaling behavior as a function of
 446 sequence length and channel count. Section 4.4 dissects the contributions
 447 of the dual-axis bidirectional design through controlled ablation. Table 5
 448 consolidates the principal quantitative results across all tasks.

Table 3: Shared training hyperparameters for SeisMamba, ResUNet, and SwinDAS. BiLSTM-DAS participates in the scalability benchmark only (Section 2.3).

Parameter	Value
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Weight decay	0.01
Initial learning rate	3×10^{-4}
LR schedule	Cosine annealing + linear warm-up (5 epochs)
Batch size	8
Max epochs	100
Early stopping patience	15 epochs
Random seeds	42, 123, 456
Hardware	$3 \times$ NVIDIA RTX 4090 (one model per GPU)
<i>Data augmentation (denoising task only)</i>	
Random time shift	± 16 samples (applied uniformly)
Channel flip	Probability 0.5
Amplitude scaling	$\mathcal{U}[0.8, 1.2]$

4.1. Denoising Performance

4.1.1. Quantitative comparison

Table 5 summarizes denoising SNR, SSIM, and Pearson correlation for SeisMamba, ResUNet, and SwinDAS, each trained with three independent random seeds. ResUNet achieves the highest denoising SNR at 29.19 ± 0.12 dB, followed by SeisMamba at 26.98 ± 0.05 dB, and SwinDAS at 19.70 ± 0.09 dB. All three models attain SSIM values above 0.98 and Pearson correlation above 0.99, indicating that the reconstructed waveforms preserve structural integrity and waveform shape across the board. The small standard deviations across seeds (0.05–0.12 dB for SNR, ≤ 0.0004 for SSIM) confirm that training is stable and that the reported means are representative rather than sensitive to weight initialization.

ResUNet’s 2.21 dB advantage over SeisMamba reflects the well-established strength of convolutional architectures for spatially local denoising tasks: residual 2-D convolutions excel at learning the compact, stationary kernels that capture the short-range spatial coherence characteristic of seismic noise patterns. SeisMamba, whose strength lies in modeling long-range sequential dependencies through selective state-space scans, achieves comparable SSIM

(0.995 vs. 0.996) and falls only marginally short in SNR, suggesting that SSM global context captures structural information nearly as effectively as local convolution while inherently processing longer contexts. The 9.49 dB gap between SwinDAS and ResUNet is more substantial and warrants interpretation. SwinDAS partitions the input into fixed local windows before applying self-attention; in these experiments the window size spans only 8 channels $\times$ 16 time samples. At this scale, coherent noise types—particularly the linear-moveout and colored noise components (see Table 2)—exhibit spatial correlations that extend well beyond a single window, preventing SwinDAS from aggregating the inter-window context needed to model and suppress such interference. This limitation is examined further in Section 5.

4.1.2. Per-noise-type breakdown

To diagnose how each architecture responds to the four noise categories in the synthetic test set, Table 4 reports per-noise-type SNR computed on non-overlapping 128×512 patches extracted from the 100 held-out test recordings.

Table 4: Per-noise-type denoising SNR (dB) on the held-out test set. Values are mean $\pm$ std over three random seeds. Bold indicates the best model for each noise type.

Noise type	SeisMamba	ResUNet	SwinDAS
Gaussian	24.88 ± 0.04	27.99 ± 0.05	20.96 ± 0.04
Colored	31.09 ± 0.10	30.48 ± 0.15	17.86 ± 0.12
Coherent	29.69 ± 0.13	33.06 ± 0.19	21.12 ± 0.20
Mixed	23.08 ± 0.04	25.53 ± 0.09	18.61 ± 0.03
Overall	26.98 ± 0.05	29.19 ± 0.12	19.70 ± 0.09

ResUNet leads in three of four categories (Gaussian, Coherent, Mixed), consistent with its overall SNR advantage from local convolutional feature extraction. The most notable result is the Colored noise category, where SeisMamba outperforms ResUNet by 0.61 dB (31.09 vs. 30.48 dB). Colored ($f^{-\alpha}$) noise has strong low-frequency spectral structure that extends across the full record; the bidirectional SSM’s global temporal context captures this long-range spectral pattern more effectively than ResUNet’s local kernels. ResUNet’s largest advantage appears on Coherent noise (33.06 vs. 29.69 dB), where its 2-D convolutional receptive field efficiently matches the spatial correlation structure of linear-moveout interference. SwinDAS lags across all categories (17.86–21.12 dB), confirming that its limited inter-window context is insufficient for any noise type in this setting.

4.1.3. Visual comparison

Figure 2 presents side-by-side panels of noisy input, clean reference, and each model’s denoised output for representative gathers covering all four noise types. Both SeisMamba and ResUNet produce denoised gathers that are visually indistinguishable from the clean references under Gaussian and colored noise, recovering the hyperbolic moveout of primary arrivals and the fine-scale wavelet character at individual receivers without introducing ringing artifacts. The most discriminating examples arise under coherent noise, where linear interference nearly masks the underlying wavefront. In those cases, SeisMamba uses its bidirectional temporal scan to identify the seismic wavefront across the full 5.12 s record and suppresses the moveout-coherent interference, while ResUNet achieves a slightly cleaner separation by exploiting local receptive fields that align well with the short spatial correlation length of the seismic arrival. SwinDAS leaves a visible residual in the coherent and colored noise cases, consistent with the quantitative SNR gap. Under mixed noise, which combines two or three noise types at reduced per-component SNR, all three models show slight degradation relative to single-type cases, but SeisMamba and ResUNet remain above 23 dB on average.

All models converge within 50–60 epochs without overfitting; SwinDAS plateaus at a higher validation loss, consistent with its limited inductive bias for long-range noise coherence.

4.2. Phase Picking Performance

4.2.1. Classification metrics

Table 5 reports three-seed mean and standard deviation for per-pixel accuracy, $F1_P$ (P-wave F1 score), and $F1_S$ (S-wave F1 score) on the held-out DeepSubDAS evaluation files. SeisMamba achieves the highest accuracy ($91.52 \pm 1.97\%$), followed by SwinDAS ($90.76 \pm 2.07\%$) and ResUNet ($90.61 \pm 2.20\%$). All three models attain $F1_S > 0.92$, reflecting the fact that S-wave segments are temporally extended and therefore well represented in the training distribution. $F1_P$ values are nominally highest for SeisMamba (0.510 ± 0.140) versus SwinDAS (0.449 ± 0.104) and ResUNet (0.446 ± 0.107); however, the inter-model spread (0.064) is less than one-half of the within-model standard deviation (~ 0.140), so these differences are *not statistically distinguishable* at any conventional significance level.

The low absolute $F1_P$ values (0.446–0.510) relative to $F1_S$ (> 0.947) reflect severe label imbalance: P-wave arrivals occupy only 4.6–22.2% of sam-

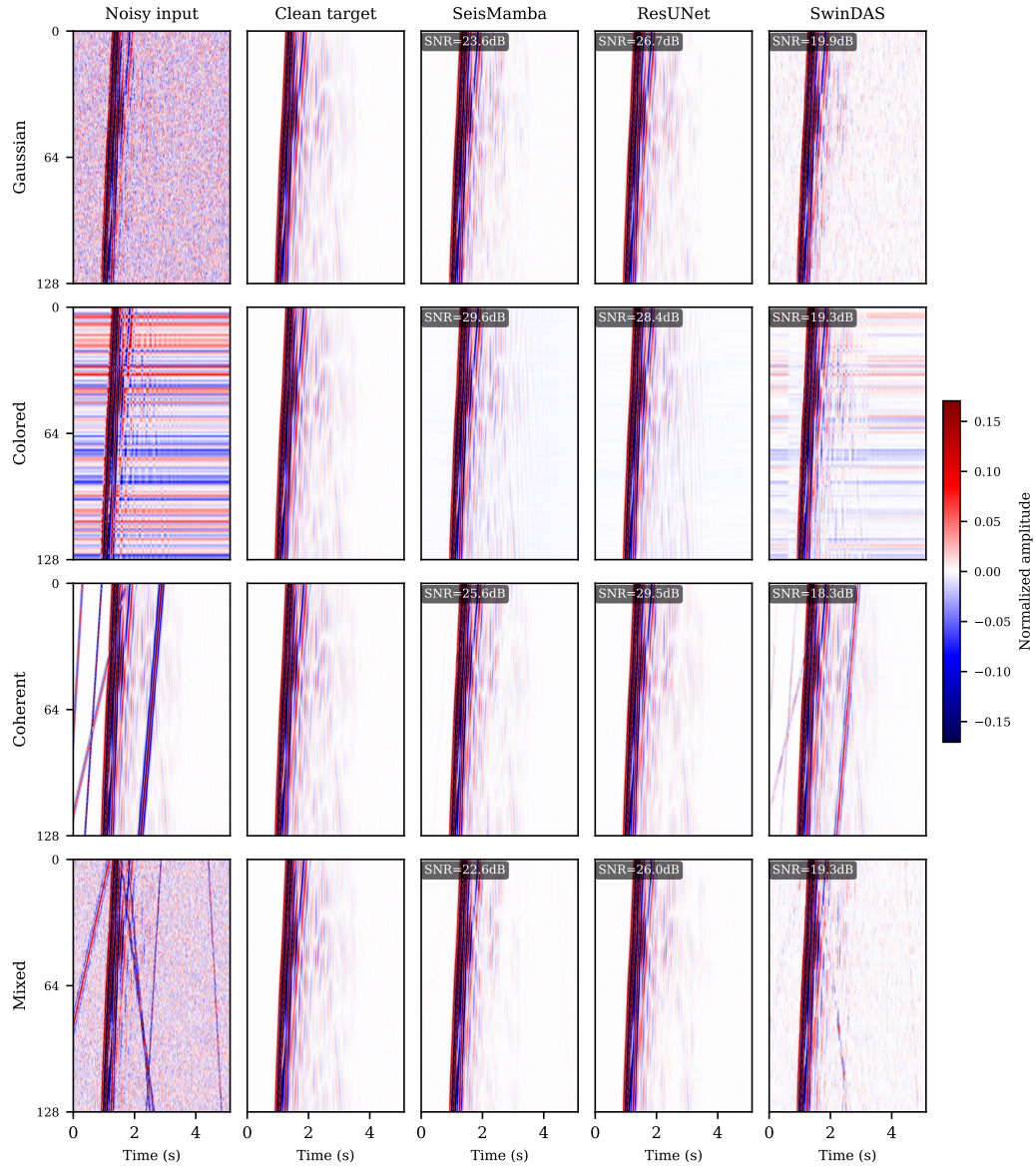


Figure 2: Denoising comparison across four noise types. Columns: noisy input, clean reference, ResUNet, SeisMamba, and SwinDAS outputs. SwinDAS retains visible residual noise under coherent conditions.

531 ples per file (median 8.3%). The narrow inter-model spread (0.064) relative
 532 to within-model seed variability (~ 0.117) confirms that $F1_P$ is a noisy metric
 533 for such sparse events.

534 4.2.2. Arrival-time precision

535 The more operationally relevant question for seismological cataloging is
 536 not whether a broad time window is labeled as P-wave, but whether the pre-
 537 dicted onset time is accurate to within a practically useful tolerance. Table 5
 538 shows the mean absolute arrival-time error MAE_t (Equation (17)) for each
 539 model: SeisMamba 0.375 ± 0.022 s, SwinDAS 0.474 ± 0.083 s, and ResUNet
 540 0.613 ± 0.052 s. SeisMamba’s MAE_t is 38.8% lower than ResUNet’s (a 0.238 s
 541 absolute improvement) and 20.9% lower than SwinDAS’s.

542 This ordering aligns with the structural differences among the architec-
 543 tures. Accurate onset localization requires the model to identify a sharp
 544 temporal boundary—the P-wave first break—against a background of noise-
 545 labeled samples. SeisMamba’s bidirectional time-axis SSM scans the full
 546 record in both directions, so the hidden state at each time step carries in-
 547 formation about what precedes and follows it; this global temporal context
 548 supports sharpening the predicted probability profile around the true onset.
 549 ResUNet processes the input with 2-D convolutions whose receptive field
 550 grows only polynomially with depth; at the moderate model size used here
 551 (four encoder stages), the global temporal context available at the output is
 552 insufficient to pinpoint an onset with the same precision [37, 36].

553 Notably, seed 456 produced the lowest $F1_P$ (0.313) yet a competitive
 554 MAE_t (0.399 s), illustrating that $F1_P$ penalizes incomplete coda labeling
 555 while MAE_t rewards precise onset localization. This reinforces MAE_t as the
 556 more diagnostic metric for practical picking applications.

557 Figure 3 illustrates predicted P-wave probability time series for three rep-
 558 resentative examples, chosen to include files with contrasting P-wave abun-
 559 dance. SeisMamba’s probability curves exhibit sharper peaks at the anno-
 560 tated onset times and lower false-alarm probability in Noise segments than
 561 either SwinDAS or ResUNet.

562 4.3. Scalability Benchmark

563 4.3.1. Sequence-length scaling

564 Figure 4 shows inference time and peak GPU memory as functions of se-
 565 quence length N_t at fixed $N_c = 256$. The four architectures exhibit qualita-
 566 tively distinct scaling regimes that match their theoretical complexity classes.

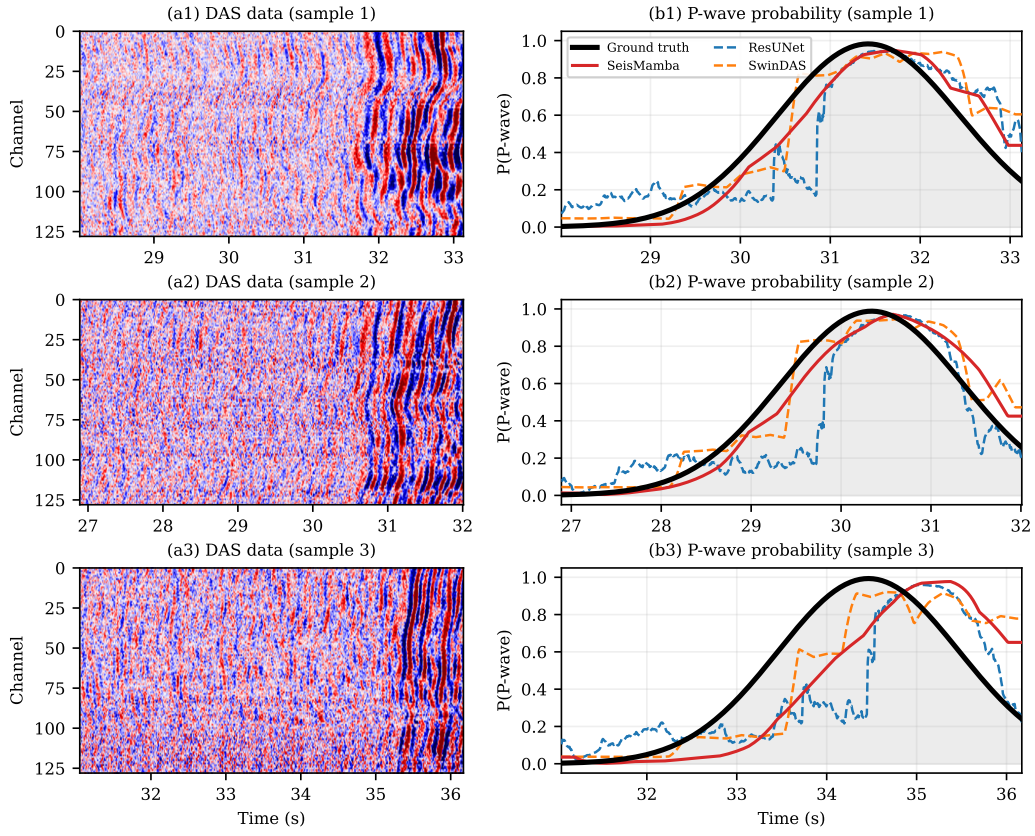


Figure 3: P-wave probability predictions on three DeepSubDAS evaluation recordings. Dashed lines mark ground-truth onsets. SeisMamba tends to produce narrower peaks closer to the annotated arrivals.

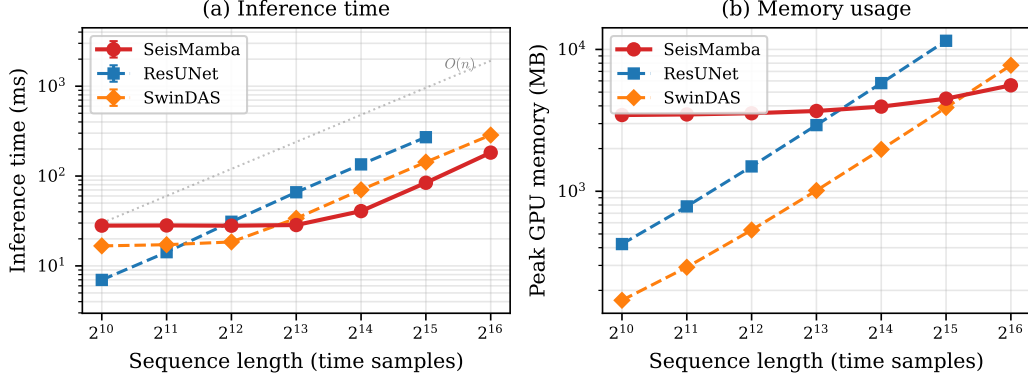


Figure 4: Scalability benchmark ($N_c = 256$ fixed). (a) Inference time vs. N_t ; SeisMamba scales linearly while BiLSTM-DAS is $\sim 10^3\times$ slower. (b) Peak GPU memory; ResUNet exceeds 24 GB at $N_t = 65,536$.

BiLSTM-DAS scales poorly in practice because the sequential hidden-state recurrence prevents full parallelization along the time axis. At $N_t = 1024$ the model requires 3,760 ms; at $N_t = 8192$ this grows to 29,365 ms—a $7.8\times$ increase for only an $8\times$ length increase, consistent with super-linear growth—and it fails to complete at $N_t = 65536$ owing to memory exhaustion. These run times are two to three orders of magnitude above the real-time processing budget for continuous 100 Hz DAS streams, confirming that BiLSTM-DAS is unsuitable for operational monitoring at scale.

ResUNet scales linearly with N_t because convolutional feature maps grow proportionally in the time dimension; however, GPU memory grows at the same rate. At $N_t = 32768$, ResUNet requires 271 ms and operates within budget, but it exhausts 24 GB of GPU memory at $N_t = 65536$ and is marked OOM. For deployments that require processing records longer than approximately 327 s at 100 Hz ($N_t \approx 32768$), ResUNet’s memory footprint becomes a practical barrier without model partitioning or chunked inference.

SwinDAS also scales linearly ($\mathcal{O}(N_t)$) because its windowed self-attention operates on local windows rather than the full sequence. Inference time rises from 17 ms at $N_t = 1024$ to 34 ms at $N_t = 8192$ (a $2.0\times$ increase for an $8\times$ length increase, reflecting efficient CUDA tiling at short lengths), then grows linearly to 143 ms at $N_t = 32768$ and 286 ms at $N_t = 65536$. SwinDAS completes all tested sequence lengths without memory errors.

SeisMamba also scales as $\mathcal{O}(N_t)$, a property inherited from the parallel selective scan algorithm that processes the full sequence in a single hardware-

efficient pass [5]. Inference time remains essentially constant at approximately 28 ms from $N_t = 1024$ through $N_t = 8192$ —a regime in which the fixed kernel launch and memory-transaction overhead dominates the linear scan cost. Beyond $N_t = 8192$, scaling follows closely linear: 84 ms at $N_t = 32768$ and 182 ms at $N_t = 65536$. At the largest tested length, SeisMamba is $1.57\times$ faster than SwinDAS (182 ms vs. 286 ms) despite both architectures belonging to the same $\mathcal{O}(N_t)$ complexity class; the absolute advantage arises from the lower per-token constant of the SSM scan kernel relative to the attention plus patch-merging operations in SwinDAS. Against BiLSTM-DAS at $N_t = 8192$, SeisMamba is $1,050\times$ faster (28 ms vs. 29,365 ms), a gap that grows further with sequence length.

Peak GPU memory profiles corroborate these trends: ResUNet’s peak allocation climbs steeply and crosses the 24 GB limit between $N_t = 32768$ and $N_t = 65536$, while SeisMamba requires 5.5 GB and SwinDAS 7.6 GB at $N_t = 65,536$ —both well within the 24 GB GPU budget.

A complementary channel-count sweep at fixed $N_t = 512$ confirms that SeisMamba’s linear scaling extends to $N_c = 8,192$ channels—a full-scale DAS array [26]—while BiLSTM-DAS and ResUNet encounter the same practical barriers observed in the sequence-length sweep.

4.4. Ablation Study

To quantify the contribution of each design choice in SeisMamba, three reduced variants were trained alongside the full model: (1) *Forward-only*, in which the bidirectional scan is replaced by a unidirectional forward scan; (2) *Channel-only*, in which only the channel-axis SSM is active; and (3) *Time-only*, in which only the time-axis SSM is active. All variants share the same parameter budget as the full Dual-Bi model and are evaluated at seed 42 for consistency.

Results are shown in Figure 5 and summarized in Table 5.

The full Dual-Bi model achieves 24.49 dB denoising SNR and 88.8% picking accuracy. Removing bidirectionality (Forward-only) decreases denoising SNR by 1.01 dB and picking accuracy by 1.9 percentage points, confirming that the backward pass supplies causally unavailable future context that is useful for both waveform reconstruction and onset detection.

Removing the time-axis SSM entirely (Channel-only) incurs a denoising penalty of 1.03 dB and a picking accuracy loss of 2.4 percentage points. Removing the channel-axis SSM (Time-only) incurs a denoising penalty of

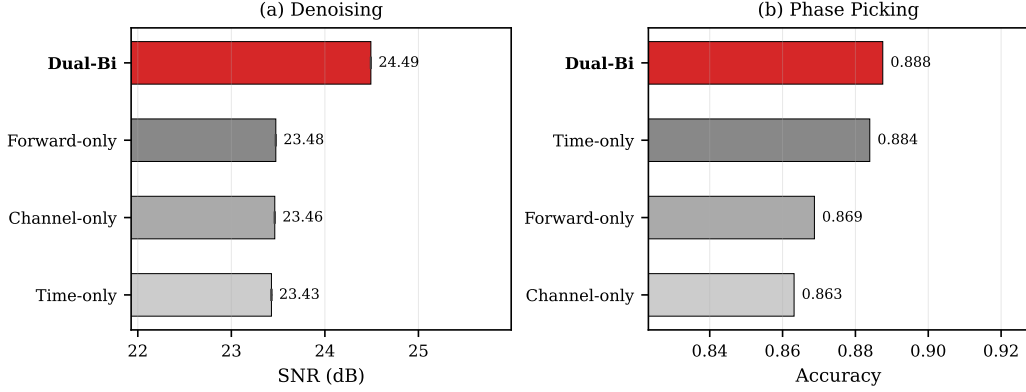


Figure 5: Ablation study (seed 42). (a) Denoising SNR and (b) picking accuracy for four SeisMamba variants. The full dual-axis bidirectional design (Dual-Bi) performs best on both tasks.

1.06 dB but reduces picking accuracy by only 0.4 percentage points. These results reveal a task-dependent interaction between the two scanning axes.

For *denoising*, both axes contribute almost equally: the 0.03 dB difference between Channel-only and Time-only lies within the range of seed-to-seed variability. Noise suppression is a 2-D problem that benefits from context along both the spatial (channel) and temporal dimensions, so suppressing either axis produces a comparable fidelity loss.

Inspection of the learned fusion weights α (channel axis) and β (time axis) across all DualMamba blocks corroborates these findings. In the denoising model, encoder blocks maintain near-equal weights ($\alpha \approx \beta \approx 0.50$), while decoder blocks elevate both axes ($\alpha \approx 0.60$, $\beta \approx 0.57$), reflecting the increased reliance on spatial-temporal context during waveform reconstruction. In the picking model, β consistently exceeds α by a small margin (mean $\beta = 0.52$ vs. mean $\alpha = 0.50$), confirming that the network learns to favor the time axis for onset detection. At the deepest encoder stage (stage 2), α drops to approximately 0.45 in both tasks, suggesting that channel-axis information is less discriminative at the coarsest spatial resolution where each token already spans 128 m of fiber.

For *phase picking*, the time axis is dominant. Time-only (no channel-axis SSM) nearly matches the full model (88.4% vs. 88.8%), whereas Channel-only (no time-axis SSM) degrades accuracy by 2.4 percentage points. This asymmetry is physically interpretable: a P-wave arrival is fundamentally a temporal event whose onset is a sharp state change in the time-domain signal.

649 The time-axis SSM captures this sequential transition directly, whereas the
650 channel-axis SSM integrates spatial coherence across receivers—secondary for
651 this task because the onset time is essentially constant across the 128-channel
652 patch.

Table 5: Summary of quantitative results. Values are mean $\pm$ std over three seeds. OOM = out of memory. Bold = best per metric.

Model	Denoising (synthetic)			Phase Picking (real)			Scalability ($N_c=256$)		
	SNR (dB)	SSIM	Corr	Acc (%)	$F1_P$	MAE $_t$ (s)	T_{8K} (ms)	T_{65K} (ms)	M_{65K} (GB)
ResUNet	29.19 $\pm$ 0.12	0.9963 $\pm$ 0.0001	0.9987	90.61 $\pm$ 2.20	0.446 $\pm$ 0.107	0.613 $\pm$ 0.052	66	OOM	OOM
SeisMamba	26.98 $\pm$ 0.05	0.9954 $\pm$ 0.0001	0.9978	91.52 $\pm$ 1.97	0.510 $\pm$ 0.140	0.375 $\pm$ 0.022	28	182	5.5
SwinDAS	19.70 $\pm$ 0.09	0.9820 $\pm$ 0.0004	0.9918	90.76 $\pm$ 2.07	0.449 $\pm$ 0.104	0.474 $\pm$ 0.083	34	286	7.6
<i>Ablation variants (seed 42, SeisMamba architecture)</i>									
Dual-Bi (full)	24.49	—	—	88.8%	—	—	—	—	—
Forward-only	23.48 (−1.01)	—	—	86.9% (−1.9pp)	—	—	—	—	—
Channel-only	23.46 (−1.03)	—	—	86.3% (−2.4pp)	—	—	—	—	—
Time-only	23.43 (−1.06)	—	—	88.4% (−0.4pp)	—	—	—	—	—

653 5. Discussion

654 The results raise three questions beyond the quantitative rankings: *why*
655 do the architectures succeed and fail where they do, what does scalability
656 mean operationally, and what are the honest limitations of this study?

657 5.1. Architectural Trade-offs

658 No single architecture dominates both tasks. ResUNet’s 2.21 dB denois-
659 ing advantage over SeisMamba reflects the strength of local 2-D convolution
660 for stationary, spatially compact noise patterns [2, 10]. Although both mod-
661 els employ residual learning, the SSM’s additional long-range expressiveness
662 does not translate into higher SNR for most noise types, because the dom-
663 inant noise components do not require global context to be identified. The
664 SSIM gap is only 0.001, confirming that structural fidelity is effectively equiv-
665 alent. The per-noise-type breakdown (Table 4) adds nuance: SeisMamba out-
666 performs ResUNet on colored ($f^{-\alpha}$) noise by 0.61 dB (31.09 vs. 30.48 dB),
667 where the long-range spectral structure of low-frequency ocean ambient noise
668 is better captured by the SSM’s global temporal context. ResUNet’s advan-
669 tage is largest on Gaussian noise (27.99 vs. 24.88 dB), whose uncorrelated
670 statistics are optimally matched to compact convolutional kernels.

671 The picking result inverts the ranking: SeisMamba achieves the lowest
672 MAE_t (0.375 s, 38.8% below ResUNet). P-wave onset localization is a global
673 temporal reasoning problem—the model must determine where the signal
674 regime begins by integrating evidence over the full record. SeisMamba’s
675 bidirectional time-axis SSM conditions each time step on the entire sequence,
676 sharpening the probability peak around the true onset [37, 36]. The ablation
677 confirms this: removing the time-axis SSM costs 2.4 percentage points, while
678 removing the channel-axis SSM costs only 0.4.

679 SwinDAS underperforms in denoising (9.49 dB below ResUNet) because
680 its 8×16 local window cannot aggregate the inter-window context needed
681 for moveout-coherent noise suppression. Its picking performance is more
682 competitive because the P-wave onset is a local temporal event that falls
683 within a single window [27].

684 These complementary interactions suggest a practical design principle: for
685 denoising at moderate array size, a CNN may be more suitable; for picking
686 or large-array processing, SeisMamba presents a viable alternative.

687 5.2. Operational Scalability Implications

688 Field DAS deployments routinely exceed 1,000–10,000 channels at 100–
689 1,000 Hz [14, 31], and real-time processing demands sub-second latency per
690 record segment. At $N_t = 65,536$, ResUNet exhausts 24 GB of GPU mem-
691 ory, BiLSTM-DAS fails entirely, and SwinDAS requires 286 ms. SeisMamba
692 completes the same task in 182 ms within 5.5 GB. A model 2.21 dB lower
693 in synthetic SNR but capable of processing full-scale arrays without memory
694 partitioning may, depending on the application, be more practical than one
695 that achieves higher benchmarks but cannot run at field scale. It should
696 be emphasized that this advantage is demonstrated at the architecture level;
697 production deployment would additionally require overlap-and-add inference,
698 I/O optimization, and domain adaptation.

699 5.3. Limitations and Future Work

700 Five limitations are identified that define a concrete research agenda.

701 *Acoustic-only synthetic data and limited dataset diversity.* The synthetic
702 data were generated with an acoustic solver (Deepwave) and do not model
703 S-wave propagation, elastic mode conversions, or surface-wave modes. The
704 picking model was trained on a single submarine cable deployment (DeepSub-
705 DAS). Neither dataset captures the full diversity of DAS acquisition condi-
706 tions. Incorporating elastic simulations and training on multiple deployment

707 types would reduce the domain gap and broaden applicability. Few-shot fine-
 708 tuning on labeled field subsets (e.g., 5–10 minutes of annotated data) is a
 709 high-priority extension.

710 *Single-task training and absence of pre-training..* Denoising and picking were
 711 trained independently. Multi-task learning with a shared encoder could
 712 leverage complementary supervisory signals [21]. Self-supervised pre-training
 713 on large unlabeled DAS archives [23, 26] would produce richer representa-
 714 tions; the $\mathcal{O}(N_t)$ complexity of Mamba makes such pre-training on multi-day
 715 recordings tractable.

716 *Fixed patch size and channel-mean pooling..* The model processes fixed $128 \times$
 717 512 patches, and the picking head aggregates spatial information by channel-
 718 mean pooling, which assumes near-constant arrival times across the patch.
 719 For wide-angle DAS geometries with significant moveout, a spatially resolved
 720 picking head would be required. Sliding-window inference with overlap-and-
 721 add would extend processing to continuous records.

722 *Scalability benchmark versus production deployment..* The benchmark re-
 723 ports single-batch inference times on random tensors under controlled condi-
 724 tions. Production deployment would require continuous-stream I/O, domain
 725 calibration, and engineering optimization common to all DL-based DAS sys-
 726 tems.

727 *Mamba-1 architecture..* This study uses the Mamba-1 selective scan. Mamba-
 728 2 [3], which reformulates the SSM as structured matrix multiplication, en-
 729 ables larger state dimensions with efficient hardware-parallel execution and
 730 is a natural architectural upgrade.

731 6. Conclusions

732 This study presented SeisMamba, the first application of selective state
 733 space models (Mamba/SSM) to seismic DAS denoising and phase picking.
 734 The proposed architecture was evaluated against parameter-matched coun-
 735 terparts — ResUNet (3.98 M, convolutional) and SwinDAS (4.11 M, Swin
 736 Transformer) — on two processing tasks and a computational scaling bench-
 737 mark.

738 In the denoising task with physics-based synthetic DAS data, ResUNet
 739 achieved the highest SNR (29.19 ± 0.12 dB), followed by SeisMamba ($26.98 \pm$

0.05 dB) and SwinDAS (19.70 ± 0.09 dB). The 2.2 dB gap between ResUNet and SeisMamba reflects the strength of local convolutional kernels for spatially compact noise patterns, while the near-identical SSIM values (0.996 vs. 0.995) indicate comparable structural fidelity. In the phase picking task with real submarine DAS recordings (DeepSubDAS), SeisMamba achieved the lowest arrival-time error ($\text{MAE}_t = 0.375 \pm 0.022$ s), 38.8% below ResUNet (0.613 s), suggesting that the bidirectional SSM is better suited to temporal onset localization. Classification accuracy and F1 scores were comparable across all three architectures, with no statistically significant inter-model differences in $F1_P$.

The scalability benchmark revealed a practically important distinction. SeisMamba sustained near-constant inference time of approximately 28 ms from $N_t = 1,000$ to 8,000 samples, rising to 182 ms at $N_t = 65,000$ while consuming 5.5 GB of GPU memory. By contrast, ResUNet exceeded the 24 GB memory limit beyond $N_t = 32,000$, and BiLSTM-DAS required roughly $1,000\times$ longer inference at $N_t = 8,000$. SwinDAS, despite sharing $\mathcal{O}(n)$ complexity through windowed attention, required 286 ms and 7.6 GB at the same sequence length.

Controlled ablation confirmed the contribution of both design choices. Removing either the channel-axis or time-axis Mamba branch degraded denoising SNR by 1.0–1.1 dB and picking accuracy by up to 2.4 percentage points. The time axis dominated picking performance, consistent with the temporal nature of seismic phase arrivals, while both axes contributed roughly equally to denoising.

Overall, no single architecture dominated all metrics. ResUNet remains the strongest choice for denoising at moderate array sizes, while SeisMamba offers a more favorable accuracy–scalability balance for scenarios involving long sequences or large channel counts. Future directions include self-supervised pre-training on unlabeled field archives, multi-task joint optimization, integration of the Mamba-2 formulation, and extension to elastic wave data. As fiber-optic sensor networks continue to expand in scale and deployment density, linear-complexity architectures are expected to play an increasingly important role in operational seismic monitoring.

Data Availability

The synthetic DAS dataset and trained model weights used in this study are available upon reasonable request to the corresponding author. The

776 DeepSubDAS field dataset is publicly available under a CC-Zero license.

777 **CRedit Authorship Contribution Statement**

778 **Youngjae Shin:** Conceptualization, Methodology, Software, Investiga-
779 tion, Writing – Original Draft, Writing – Review & Editing, Visualization.

780 **Declaration of Competing Interest**

781 The author declares no known competing financial interests or personal
782 relationships that could have appeared to influence the work reported in this
783 paper.

784 **Acknowledgments**

785 The author thanks the anonymous reviewers and the editor for their
786 constructive comments that improved this manuscript. The DeepSubDAS
787 dataset used for phase picking evaluation was made publicly available by
788 Yuan et al. [30]; their effort in providing open-access labeled DAS recordings
789 is gratefully acknowledged.

790 **Declaration of generative AI and AI-assisted technologies in the** 791 **manuscript preparation process**

792 During the preparation of this work the author used Claude (Anthropic)
793 in order to assist with literature search, experiment design, English language
794 editing, and grammar checking. After using this tool, the author reviewed
795 and edited the content as needed and takes full responsibility for the content
796 of the published article.

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